IS-4543: Cyber Attack and Defend

Project Milestone Report for Milestone 1 -- Mini SOC Using Wazuh Project

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Section: IS-4543-001

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Milestone Report

This report includes the approved project proposal (included for easy reference), a summary of what was done during this project milestone, a description of what I learned during this milestone, and documentation of the work completed during this milestone.

Project Proposal

Basic Mini SOC Using Wazuh

I want to use virtual machines and Wazuh to build a simple Security Operations Center (SOC) lab for this project. In my personal lab, I'll use Kali Linux to generate various security events against a system and Wazuh to track the results. I want to test things like failed login attempts and network scans. The project's main goal is to learn how a SIEM gathers logs, identifies suspicious activity, and assists someone in looking into a potential security event.

Milestone 1 - Plan the Lab

I will research what I need to set up the lab and decide how each virtual machine will be used. I will also create a network diagram showing how the systems are connected and what I plan to test.

Milestone 2 - Set up Wazuh

I will set up the virtual machines and install Wazuh. Then, I will connect the system I want to monitor to Wazuh and make sure that logs are being collected correctly before I start testing anything.

Milestone 3 - Create Security Events

I will use Kali Linux to create basic security events inside my own lab. This will include things like scanning the target system and attempting failed logins. I will also document the commands I use and what happens during each test.

Milestone 4 - Review the Alerts

I will look at the alerts that Wazuh creates from my tests and figure out what information they provide and look at things such as where the activity came from, what system was affected, and what caused the alert.

Milestone 5 – Final Testing and Results

I will repeat some of my tests to make sure Wazuh is detecting the activity correctly and will organize my screenshots, alerts, network diagram, and results and explain what I learned from building and testing the lab.

Summary of Activities (5 points)

For my first milestone, I focused on planning out my Mini SOC lab before installing anything. I wanted to make sure my laptop could handle the virtual machines and that I had a clear idea of how everything would connect. The three main systems I chose were Kali Linux, Ubuntu Linux, and Wazuh. Kali will be used to generate security events, Ubuntu will be the system being monitored, and Wazuh will collect and analyze the logs.

I started by checking my laptop's hardware through Windows Task Manager and File Explorer. My laptop has 32 GB of RAM, an Intel Core i7-12800H processor with 14 cores and 20 logical processors, and about 205 GB of available storage. I also confirmed that virtualization was enabled. After checking the hardware, I planned how much RAM, CPU, and storage to give each virtual machine. I decided on 2 GB of RAM for Kali, 2 GB for Ubuntu, and 8 GB for Wazuh. I originally planned to give Wazuh 4 GB, but increased it to 8 GB after looking into what it needs to run its monitoring components.

Table 1. *Planned virtual machine configuration*

System	Role	Planned IP	RAM	vCPUs	Disk
Kali Linux	Attacker / test activity	192.168.159.10	2 GB	2	30 GB
Ubuntu Linux	Monitored endpoint (Wazuh agent)	192.168.159.20	2 GB	2	25 GB
Wazuh Server	SIEM / security monitoring	192.168.159.30	8 GB	2	50 GB
Total			12 GB	6	105 GB

These are my starting numbers, and I may change them once everything is installed.

Next, I opened VMware Workstation's Virtual Network Editor to check how I could keep the lab isolated. I found that VMnet1 was already configured as a host-only network using the 192.168.159.0/24 subnet. I also checked the DHCP settings and found that VMware assigns IP addresses from 192.168.159.128 through 192.168.159.254. Based on that information, I chose three planned static IP addresses: 192.168.159.10 for Kali, 192.168.159.20 for Ubuntu, and 192.168.159.30 for Wazuh. These addresses are outside the DHCP range, although I'll still need to verify that they're available before assigning them.

After figuring out the network settings, I used draw.io to create my network diagram. I started with three basic boxes and then updated the diagram to include each machine's role, planned IP address, RAM, CPU, and storage. I also added arrows showing the security tests I plan to run from Kali to Ubuntu and how Ubuntu will send security logs to Wazuh. Finally, I added a dashed boundary to show that all three machines will be connected through the same host-only network.

By the end of this milestone, I had finished checking my hardware, planning the virtual machine resources, reviewing VMware's network settings, and creating the network diagram. I saved screenshots of my work along the way so I could document the planning process. I haven't installed or configured the virtual machines yet, since that will be part of my second milestone.

Description of Learning Completed (5 points)

One of the biggest things I learned from this milestone was how important it is to plan a lab before starting the installation process. I've worked with virtual machines before, but I hadn't really thought about how much RAM, CPU, and storage a setup like this would need. At first, I planned to give the Wazuh server 4 GB of RAM, but after looking into its requirements, I realized that might not be enough for the full setup. I changed it to 8 GB because Wazuh has multiple components running together, including the manager, indexer, and dashboard.

I also learned that having 32 GB of RAM doesn't mean I can give all of it to my virtual machines. Windows and other applications are already using some of that memory. If I allocate too much RAM to the VMs, my laptop could slow down, especially when all three machines are running. This made me realize that I need to pay attention to the available resources, not just the total amount installed.

Another thing I learned was the difference between host-only and NAT networking in VMware. A host-only network allows virtual machines to communicate with each other and the host computer without giving them direct internet access. NAT networking allows virtual machines to access outside networks through the host. For this project, I chose host-only networking because I plan to run network scans and failed login attempts inside my own lab. I don't want that activity reaching other devices on my regular network. I also learned that I may need to temporarily use NAT when downloading updates or installing software, then switch back to the isolated network for testing.

Checking the DHCP settings was another useful part of this milestone. Before this, I hadn't really considered how VMware automatically assigns IP addresses to virtual machines. I found that my DHCP range starts at 192.168.159.128 and ends at 192.168.159.254. This helped me understand why choosing static IP addresses outside that range can make the lab easier to manage. For example, Ubuntu's Wazuh agent will need to know the Wazuh server's address. Keeping that address consistent should help avoid connection problems later.

Lastly, creating the network diagram helped me understand how the three systems will work together. Kali will generate the test activity, Ubuntu will produce logs from that activity, and Wazuh will help me review the security events. I also learned that a network diagram should show more than just the machines being used. Adding the IP addresses, system roles, and connections makes it easier for someone else to understand the setup. The diagram is still a plan, so I'll use it during the next milestone to check whether everything is connected and working the way I intended.

Documentation of Work Completed (5 points)

These are the screenshots I took while checking my hardware and VMware settings, plus my final network diagram. Nothing has been installed yet.

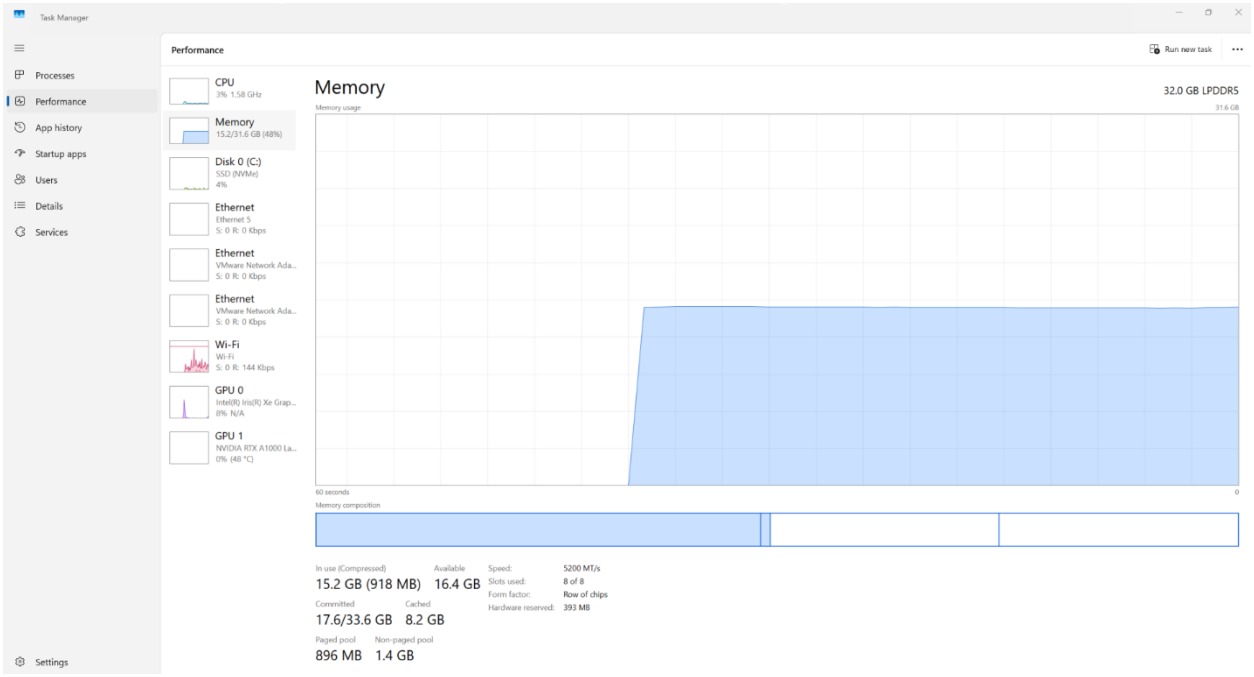


Figure 1: Windows Task Manager showing 32 GB of installed memory.

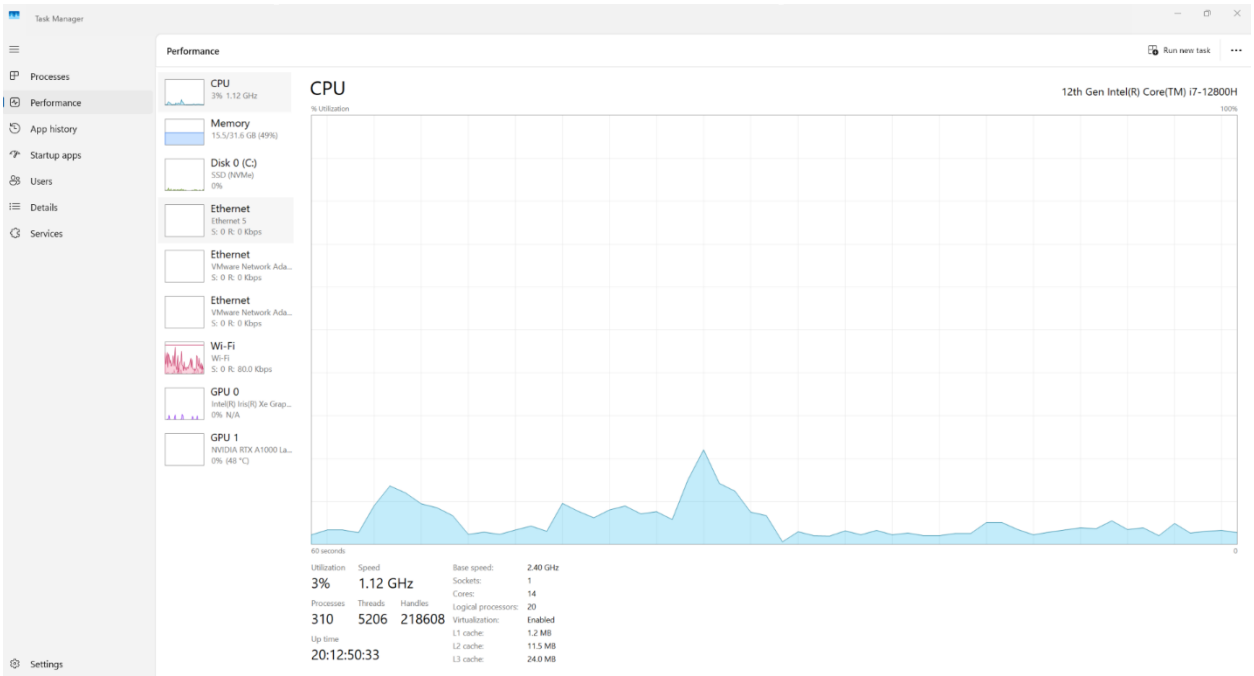


Figure 2: Windows Task Manager showing the Intel Core i7-12800H processor and virtualization enabled.

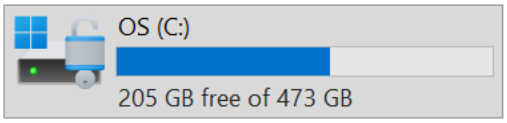


Figure 3: File Explorer showing 205 GB free out of 473 GB on the C: drive.

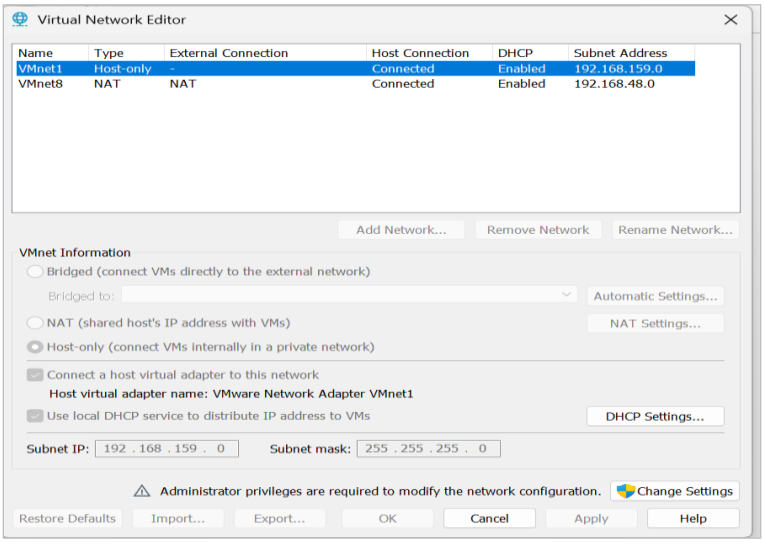


Figure 4: VMware Virtual Network Editor showing the VMnet1 host-only subnet (192.168.159.0/24) and VMnet8 NAT subnet.

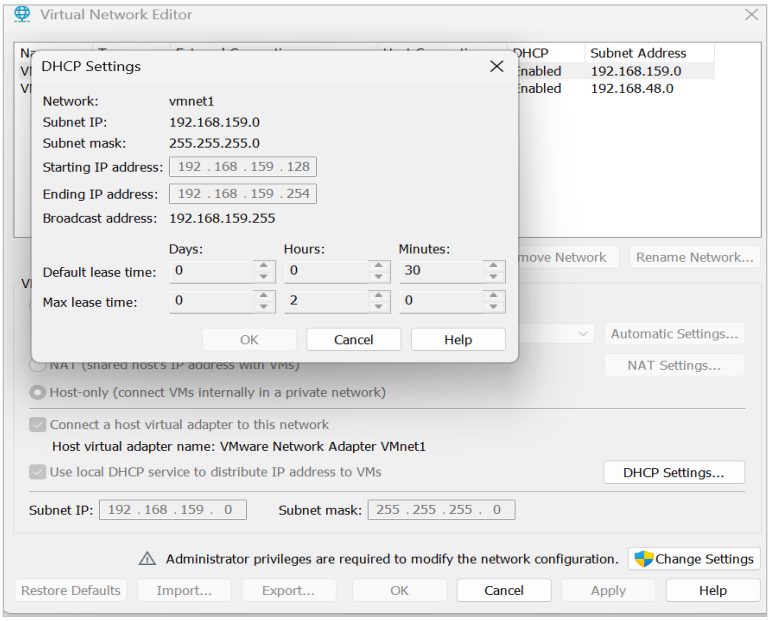


Figure 5: VMnet1 DHCP settings showing the dynamic range 192.168.159.128–.254, used to choose non-conflicting static IP addresses.

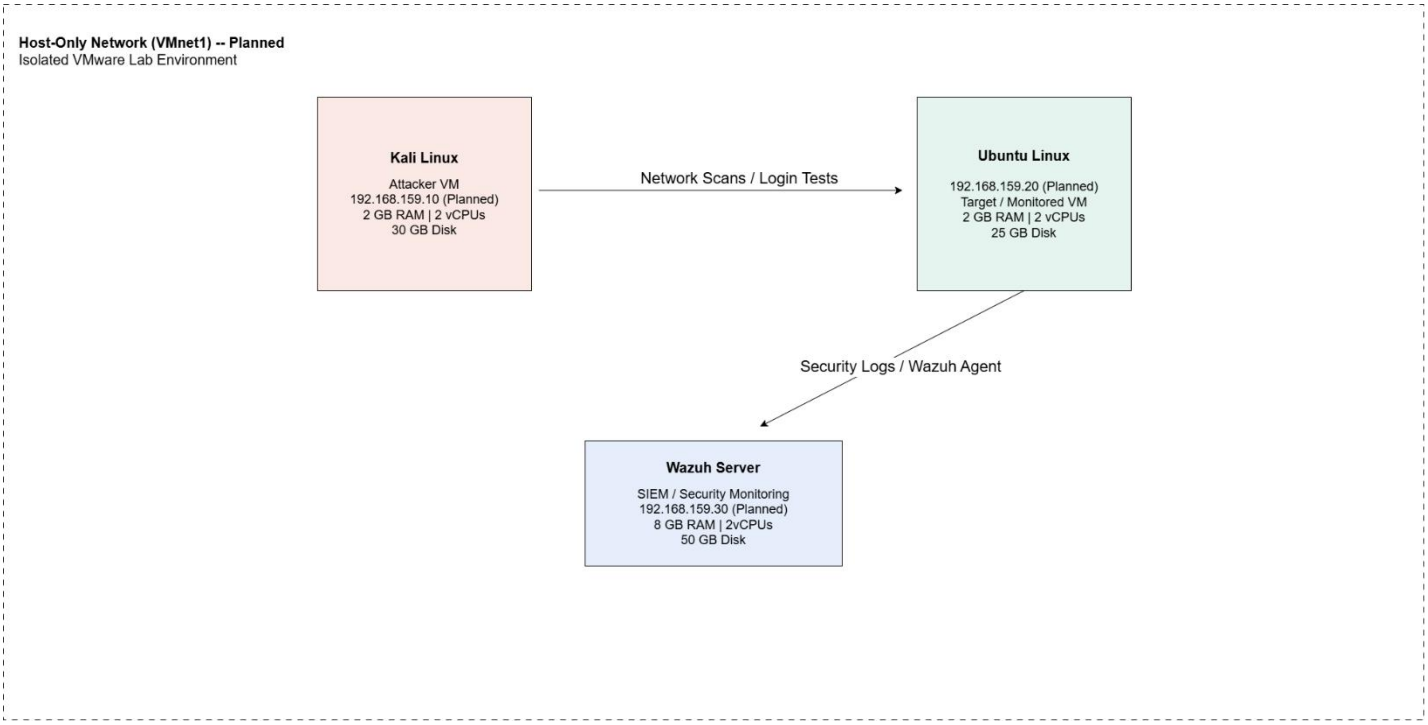


Figure 6: Planned Mini SOC architecture showing testing activity, monitored endpoint logs, and the Wazuh server.

Next Steps

For Milestone 2, I will create the three VMs in VMware, set the static IPs on the host-only network, and install the Wazuh server. Then I will install the Wazuh agent on Ubuntu, connect it to the server, and make sure logs are coming in before I run any tests. I will take screenshots and write down any problems I run into and how I fix them.

References

Broadcom. (n.d.). VMware Workstation Pro documentation. https://knowledge.broadcom.com/

Canonical. (n.d.). Ubuntu Server documentation. https://documentation.ubuntu.com/server/

diagrams.net. (n.d.). draw.io diagramming application. https://app.diagrams.net/

Kali Linux. (n.d.). Kali Linux documentation. https://www.kali.org/docs/

Wazuh. (n.d.). Wazuh documentation. https://documentation.wazuh.com/